

A Literature-Implied Positive Answer on Complex Radon Planes

Source arXiv:2108.00655; supporting construction arXiv:2402.13416

Automated mathematics run `fa_banach_001`, lane 5

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Status: literature-implied answer. The later paper constructs a nonsmooth complex Radon plane. It cites the source article elsewhere, but does not explicitly frame the construction as answering the source sentence, so the implication is recorded rather than an explicit literature answer.

Source question

The source is R. Tanaka, *On Birkhoff–James orthogonality preservers between real non-isometric Banach spaces*, arXiv:2108.00655 [1]. In the Remarks on PDF page 11, the author explains that the complex analogue of the construction would require non-Hilbert Radon planes and states that, to the best of his knowledge, it was not known whether a complex Radon plane exists. Since every two-dimensional complex Hilbert space has symmetric Birkhoff–James orthogonality, the surrounding sentence makes clear that the intended issue is a *non-Hilbert* complex Radon plane.

Recall that a two-dimensional normed space is a Radon plane when its Birkhoff–James orthogonality is symmetric.

Supporting construction

The supporting paper is A. Guterman, B. Kuzma, S. Singla, and S. Zhilina, *Birkhoff–James classification of norm’s properties*, arXiv:2402.13416 [2]. Example 3.17 on PDF pages 23–24 gives an explicit positive answer.

Start with the absolute nonsmooth real Radon plane $(\mathbb{R}^2, \|\cdot\|_{\mathcal{R}})$ constructed in Example 3.14 and put

$$\mathcal{X} = \mathbb{C}^2, \quad \|(a, b)\|_{\mathcal{X}} = \||a|, |b|\|_{\mathcal{R}}.$$

For $(c, d) \in \mathbb{C}^2$, identify the functional $\widehat{(-d, c)}$ by

$$\widehat{(-d, c)}(a, b) = -da + cb.$$

The computation in Example 3.17 proves

$$\|\widehat{(-d, c)}\|_{\mathcal{X}^*} = \|(c, d)\|_{\mathcal{X}}.$$

Thus

$$\Phi(c, d) = \widehat{(-d, c)}$$

is an isometry from $\mathcal{X}$ to $\mathcal{X}^*$ and satisfies $\Phi(x)(x) = 0$ for every $x \in \mathcal{X}$. By Lemma 3.15 of the supporting paper, this property is equivalent in dimension two to being a Radon plane. Hence $\mathcal{X}$ is a complex Radon plane.

The same example shows that $(1, 0)$ is a nonsmooth point of $\mathcal{X}$. Therefore $\mathcal{X}$ is not a Hilbert space, giving exactly the non-Hilbert example required by the source context. The supporting authors additionally note that earlier complex Radon-plane examples already appeared in Oman's work; their Example 3.17 is nonetheless a self-contained decisive source for the present identification.

Provenance and scope

The supporting paper cites Tanaka's source article for a different orthogonality-preserver construction, but does not explicitly label Example 3.17 as an answer to Tanaka's Remarks. The relation is therefore an agent-identified direct implication and belongs in `literature_implied_answers`.

This packet settles only existence of a non-Hilbert complex Radon plane. It does not settle the source's automatic-continuity question for preservers between smooth complex Banach spaces, the nonreflexive and finite-dimensional gaps in that classification, or the unrestricted implication $X \sim_{BJ} Y \Rightarrow X \cong Y$ (and its complex conjugate variant).

References

- [1] R. Tanaka, *On Birkhoff–James orthogonality preservers between real non-isometric Banach spaces*, Indag. Math. (N.S.) **33** (2022), no. 6, 1125–1136; arXiv:2108.00655.
- [2] A. Guterman, B. Kuzma, S. Singla, and S. Zhilina, *Birkhoff–James classification of norm's properties*, arXiv:2402.13416, Example 3.17, pp. 23–24.